Trachoma

Trachoma, or **granular conjunctivitis**, is a common cause of preventable blindness that is rare in the United States but widespread in developing countries, especially in Africa and Asia. The condition is caused by the same species that causes neonatal inclusion conjunctivitis in infants, *Chlamydia trachomatis*. *Chlamydia trachomatis* can be transmitted easily through fomites such as contaminated towels, bed linens, and clothing and also by direct contact with infected individuals. *Chlamydia trachomatis* can also spread by flies that transfer infected mucous containing *Chlamydia trachomatis* from one human to another. Infections of eye are listed in Table 12.6.

12.6 Urinary Tract Infections

The urinary system is composed of organs that regulate the chemical composition and the volume of the blood excrete mostly nitrogenous wastes products and water.

Infobits

Many of the bacteria which cause UTI's have developed resistance to antibiotics. Research has turned to probiotic (*Lactobacillus*) strain which stimulates immune function, lowers acidity levels in the urinary tract, and discourages the growth of UTI causing organisms.

The urinary system consists of two kidneys, two ureters, a single urinary bladder and a single urethra. Wastes are removed from the blood as it circulates through the kidneys (Figure 12.15).

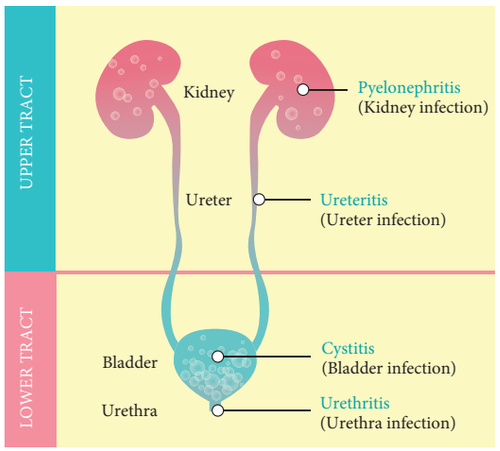


Figure 12.15: Structure of lower and upper urinary tract infection

Infections of the kidney, ureter and bladder constitute Urinary Tract Infections (UTI). When infection occur in the kidney and ureter it is called upper urinary tract infections and bladder downwards is called lower urinary tract infections. Urinary tract infection is common in females than males. The urinary system normally contains few microbes but it is subjected to opportunistic infections that can be quite troublesome. Almost all such infections are caused by bacteria although occasional infections by pathogens such as parasites, protozoa and fungi also occured. Microorganisms invloved in UTI are listed in Table 12.7.

Table 12.7: Microorganisms involved in UTI

Microorganisms	Examples
Bacteria (most common)	*Escherichi coli, Klebsiella, Enterobacter, Proteus*
Viruses	Adenovirus, Mumps
Fungi	*Trichomonas vaginalis, Schistosoma haematobium*
Parasites	*Candida*

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